

curriculum vitae

Nuclear Materials Modeling Specialist with a Ph.D. in Computational and Experimental Mechanics of Materials and over a decade of experience developing advanced numerical-experimental metrology methods and physics-based constitutive models for nuclear reactor materials under extreme conditions. Expertise in finite element analysis, Python/MATLAB algorithm design, and multi-scale modeling with verification and validation, and large-scale uncertainty quantification. Combines deep mechanics of materials knowledge with programming skills to deliver innovative computational solutions for nuclear systems. Effective communicator who bridges detailed scientific analysis with real-world application needs in multidisciplinary teams.

Personalia

Surname:	Ruybalid
Given names	Andre, Paul
Degrees:	Ph.D. MSc. BSc.
City:	Los Alamos, New Mexico, U.S.A.
Phone number:	+1 719 210 1746 (personal mobile)
E-mail:	andreruybalid@gmail.com
Date of birth:	26 October 1987
Place of Birth:	Farmington, New Mexico, U.S.A.
Nationality:	American/Dutch

Professional experience

2023 - Present

R&D Engineer, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed multiple surrogate models and custom Python-based tools to bridge polycrystalline mechanics of nuclear reactor materials to engineering-scale predictions for nuclear reactor applications under high-temperature and neutron fluence conditions.
- ❖ Built and maintained material models for nuclear fuel performance analysis in MOOSE/BISON, including code contributions, verification, validation, and uncertainty quantification. (UQ)
- ❖ Designed custom Python algorithms and data analysis pipelines to automate simulation workflows and enable engineering-scale structural predictions from thousands of low-length-scale simulations.
- ❖ Coordinated with mechanistic modelers as part of a larger multi-scale modeling workflow for nuclear reactor materials behavior.
- ❖ Reached the R&D 100 Award Finals twice with the surrogate modeling solutions for nuclear materials applications.

2021 – 2023

Postdoctoral Researcher, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

Developed the surrogate model ("LAROMance") for polycrystalline nuclear reactor materials by designing large-scale simulation pipelines and curating datasets from thousands of low-length-scale simulations using Python-based machine learning approaches and custom tooling.

2019 – 2020

Design Engineer, ASML

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and validated advanced metrology systems for high-precision nanoscale defect detection, performing specialized cleanroom testing and data analysis under tight deadlines in collaboration with cross-functional and industrial teams.

2013 – 2019

PhD Researcher, Eindhoven University of Technology

Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.

Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands

Designed and implemented an integrated computational-experimental framework combining finite element modeling using MSC Marc/Mentat (with Fortran-based user subroutines for custom constitutive models) alongside MATLAB/Python algorithm development and inverse optimization techniques to characterize material system behavior, resulting in five journal publications.

2017 – 2020

Account manager & Technical recruiter

Research-Square & EuFlex | Eindhoven, The Netherlands

Acquired and managed client accounts, and conducted technical recruitment of highly specialized engineering and PhD-level candidates for science and technology companies.

2018 – 2019

Lab technologist, Multi-Scale Laboratory

Eindhoven University of Technology | Eindhoven, The Netherlands

Computational and Experimental Mechanics

Managed an experimental mechanics laboratory, overseeing high-tech equipment including scanning electron microscopes and micro-mechanical test systems, while providing technical guidance to researchers and students and supporting experimental design in a high-demand environment.

Education

Eindhoven University of Technology, the Netherlands

Department of Mechanical Engineering

Division: Computational and Experimental Mechanics

Research Group: Mechanics of Materials

2013 – 2019

PhD.-research project

Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."

2013 – 2019

Graduate School of Engineering Mechanics

Advanced Mechanics Courses

2010 – 2013

MSc. Mechanical Engineering

Minor: Biomedical Engineering

2006 – 2010

BSc. Mechanical Engineering

2000 – 2006

Research-oriented secondary education (VWO)

St.-Janscollege, Hoensbroek, The Netherlands

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). <https://doi.org/10.1016/j.commatsci.2025.114367>.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. Integrating Materials and Manufacturing Innovation 13, 895–914 (2024). <https://doi.org/10.1007/s40192-024-00377-z>

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), <https://doi.org/10.1016/j.engfracmech.2020.106879>.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), <https://doi.org/10.1007/s11340-020-00583-8>.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), <https://doi.org/10.1016/j.ijsolstr.2018.08.010>.